

EMILIA "EMI" RUTH

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SUMMARY

Information security engineer with three years experience at a publicly traded software company. Experience with zero-trust infrastructure deployment and operation, detection tuning, incident response, PKI and HSM operations, and hands-on evaluation of emerging security products.

PTC Information Security Engineer

Boston MA | January 2025 – Present

- Own engineering and administration of the company's Zscaler deployment (ZIA and ZPA) covering 5,000+ employees, 9,000+ enrolled devices, partner-tenant and browser-based customer access, and two federated Entra ID tenants.
- Served as lead engineer on recent endpoint detection and response product proof of concept "bake off," running integration and deployment, product testing, and reporting to CISO and senior leadership on product impressions.
- Evaluate new security products as a regular technical evaluator and early integration staff member. Recent work spans AI security monitoring (endpoint agent, browser extension, MCP gateway), cyber resilience, endpoint detection and response, and software supply chain firewalling.
- Increased TLS inspection coverage from 15% to 48% of enterprise traffic by auditing roughly 800 legacy URL-category bypasses, building a re-categorization override process, and removing redundant rules — substantially expanding detection visibility with no user-facing disruption.
- Reduced open alerts on an AI security monitoring platform from 123,000 to 17,000 (86%) by designing runtime rules that suppress sanctioned enterprise AI application activity while preserving auditable records of every event.
- Administer corporate PKI, migrating the offline root CA to new hardware, renewing the issuing CA, and executing an emergency root CRL republication ahead of a new issuing CA rollout; supported migration of Thales Luna 7 HSMs to Data Protection on Demand.
- Built Python and JavaScript tooling against Zscaler APIs for one-time-password generation, device inventory, and reporting; designed granular API roles to reduce standing privilege.
- Stood up a working ZPA disaster recovery environment and authored the production runbook; currently leading DR design strategy
- Identified an internet-exposed AWS resource permitting anonymous authentication, traced the owning team, and drove remediation of a stalled finding.
- Responded to a Severity 1 ZPA infrastructure outage in Azure, personally restoring five servers; separately executed emergency removal and same-night restoration of guest network access.
- Regularly present change proposals to the IT Change Advisory Board; author written analyses and knowledge base articles; assist in user support for specialized ZPA and ZIA networking issues.

Rotational Leadership Development Program

July 2023 – December 2024

- Joined as a mechanical engineering graduate and completed three six-month rotations under the VP of Atlas Platform Product Management, the Chief Information Officer, and the Chief Information Security Officer. Recruited directly by the CISO into a permanent security engineering role at the program's conclusion.
- Contributed to migration from traditional corporate network and VPN to a zero trust and café-model architecture.
- Co-owned enterprise migration from several productivity software providers to a unified Miro platform, preserving all user documents and assets.

EDUCATION

Brown University — Sc.B., Mechanical Engineering

Providence, RI | 2023

Relevant coursework: Computers, Freedom and Privacy (CSCI 1805); Introduction to Object-Oriented Programming (CS 0150); Computer Aided Design (ENGN 1740)

Varsity Athletics: Sailing, women's crew (2018–2023) — Second Team All-Ivy 2023, ICSCA Academic All-American 2021 and 2022. Alpine Skiing (2018–2020) — USCSA Regionals and Nationals qualifier. Three-season varsity athlete, 2018–2020.

SKILLS & CERTIFICATIONS

Security: Zero trust architecture (Zscaler ZIA/ZPA/ZDX/ZCC), TLS inspection, PKI and certificate lifecycle, HSM operations (Thales Luna, DPOD), privileged access management (Delinea Secret Server), DLP (Microsoft Purview), EDR (CrowdStrike, Microsoft Defender), SIEM and log analysis (Splunk), incident response

Cloud & Platforms: Microsoft Azure, Microsoft Entra ID, AWS (working knowledge), Docker, ServiceNow

Languages & Automation: Python, JavaScript, Bash/zsh, SQL, REST API automation, Ansible

Certifications: CompTIA Security+ (2025), Zscaler Delivery Specialist, Udacity Full Stack Developer Nanodegree